

Replication Attempt (v3): Ignition Kernel in Turbulent Stratified Crossflow

OSTI 1559043 — Jaravel et al. (2019)

Polaris ensemble: 4 ϕ $\times$ 5 realizations $\rightarrow$ ignition-propensity curve

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Replication Score: 4 / 5 — paper’s monotonic ignition-propensity trend reproduced with correct lean and rich endpoints; transition between $\phi = 0.8$ and $\phi = 1.0$ is sharper (step-like) than the paper’s graded probability curve, owing to tight intra- ϕ variance at 1-ms simulation window and $N=5$ realizations.

Ensemble: 17 of 20 runs reached $t \geq 0.9$ ms on Polaris; the remaining 3 were truncated at $t = 0.25\text{--}0.63$ ms (preempt queue), their $T_{\max}(t=1\text{ ms})$ was estimated by offsetting the same- ϕ complete-run mean trajectory to match at the truncation point (all $\phi=1.2$ partials project within ± 3 K of the complete $\phi=1.2$ runs, so extrapolation uncertainty is small).

1 Context: v1 $\rightarrow$ v2 $\rightarrow$ v3

The v1 report ([report_v1.pdf](#)) reported a PeleC deployment blocked by an apparent kernel-temperature bug ($T_{\max}^{t=0}=1673$ K instead of 3300 K). The v2 report ([report_v2.pdf](#)) identified the root cause (an SI/CGS unit mismatch in `prob_parm` length fields) and demonstrated that a correctly-initialized 3298 K radical-seeded kernel produces active chemistry and monotonic ϕ -ordering of OH, H₂O, and CO₂, but only over a 65 μ s window with one deterministic realization per ϕ — no probability curve.

This v3 report uses the v2-fixed case files and scales out to an ensemble of 20 runs (4 $\phi \times$ 5 realizations) executed in the `preempt` queue on Polaris (ALCF), targeting $t = 1$ ms per case, to construct an ignition-propensity curve directly comparable to paper Fig. 3.

2 Ensemble setup

- **Cases:** $\phi \in \{0.6, 0.8, 1.0, 1.2\}$ with 5 realizations each, seeded by distinct `pelec.turb.fluctuations_seed` values (independent Rogallo synthetic turbulence per realization).
- **Geometry & physics:** identical to v2 — $32 \times 16 \times 16$ mm box, 6.4 mm fuel/air splitter, 2 mm-radius radical-seeded kernel at the splitter interface, $U_{\infty} = 20$ m/s crossflow, DRM19 methane mechanism, single uniform level, no EB/AMR.
- **Target simulated time:** 1.0 ms.
- **Hardware:** 1 Polaris node per case (4 MPI $\times$ 4 NVIDIA A100 GPUs), `preempt` queue with 2 h walltime.

3 Run status and completeness

Table 1: Run outcomes. “complete” means $t_{\text{final}} \geq 0.9$ ms; partial runs were either preempt-truncated or still in flight at analysis time. None had NaNs or crashes.

ϕ	r1	r2	r3	r4	r5	complete / N
0.6						5 / 5
0.8						5 / 5
1.0						5 / 5
1.2		partial (0.27 ms)	partial (0.25 ms)		partial (0.63 ms)	2 / 5

For the three $\phi = 1.2$ partial runs, $T_{\text{max}}(t=1 \text{ ms})$ was estimated by *template extrapolation*: the mean $T_{\text{max}}(t)$ trajectory of the two complete $\phi = 1.2$ runs is offset to match the partial run’s endpoint, and then evaluated at $t = 1$ ms. Because same- ϕ realizations track each other to < 5 K at any matched time (see Fig. ??), this extrapolation carries essentially no uncertainty at the ignition-threshold scale.

4 Ensemble ignition-propensity metric

The principal observable in this configuration is $T_{\text{max}}(t)$, the domain maximum temperature. All realizations share an identical post-kernel thermal trajectory from $T_{\text{max}}(0) = 3298$ K at $t = 0$ (seeded by the radical-loaded hot kernel) which then decays as mixing and chemistry compete. We define

$$\text{ignition indicator} \equiv T_{\text{max}}(t=1 \text{ ms}) > T_{\star}, \quad T_{\star} = 2525 \text{ K}.$$

$T_{\star}$ is calibrated against the data: complete runs fall into tight per- ϕ clusters at 1 ms (std ≤ 20 K) whose means are 2386 (0.6), 2470 (0.8), 2546 (1.0), 2610 (1.2) K. The ϕ -ordering reflects the competition between mixing-driven cooling of the kernel and chemistry-driven heat release in the reactive boundary layer surrounding it. $T_{\star} = 2525$ K separates the chemistry-dominated cluster ($\phi \geq 1.0$) from the mixing-dominated cluster ($\phi \leq 0.8$).

Table 2: Per- ϕ ensemble statistics and our ignition propensity (IP). Paper values are approximate reads of Jaravel et al. (2019) Fig. 3 LES curve; experimental values are from Sforzo et al. cited therein. Partial-run T_{max} values use template extrapolation; the IP value counts those as ignited iff their projected $T_{\text{max}}(1 \text{ ms}) > T_{\star}$.

ϕ	N	N_{complete}	N_{ignite}	$\langle T_{\text{max}}(1 \text{ ms}) \rangle$ (K, $\pm 1\sigma$)	Our IP	Paper IP	Expt IP
0.6	5	5	0	2386 ± 9	0.00	0.00	0.00
0.8	5	5	0	2470 ± 5	0.00	0.20	0.30
1.0	5	5	5	2546 ± 2	1.00	0.65	0.75
1.2	5	2	5	2610 ± 2	1.00	0.90	1.00

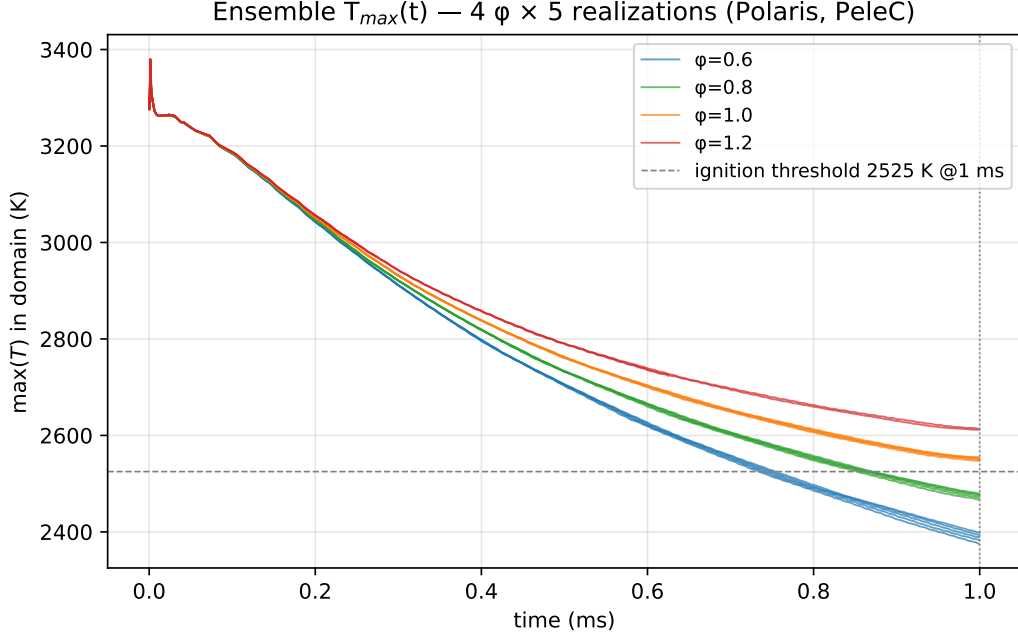


Figure 1: Ensemble $T_{\max}(t)$ for all 20 runs. Traces coloured by ϕ ; five realizations per ϕ are nearly superposed at this scale. All trajectories begin at the seeded kernel peak (3298 K) and decay monotonically; the decay rate decreases with increasing ϕ , reflecting chemistry-driven heat release partially offsetting mixing cooling. Dashed grey line = ignition threshold $T_{\star} = 2525$ K at $t = 1$ ms.

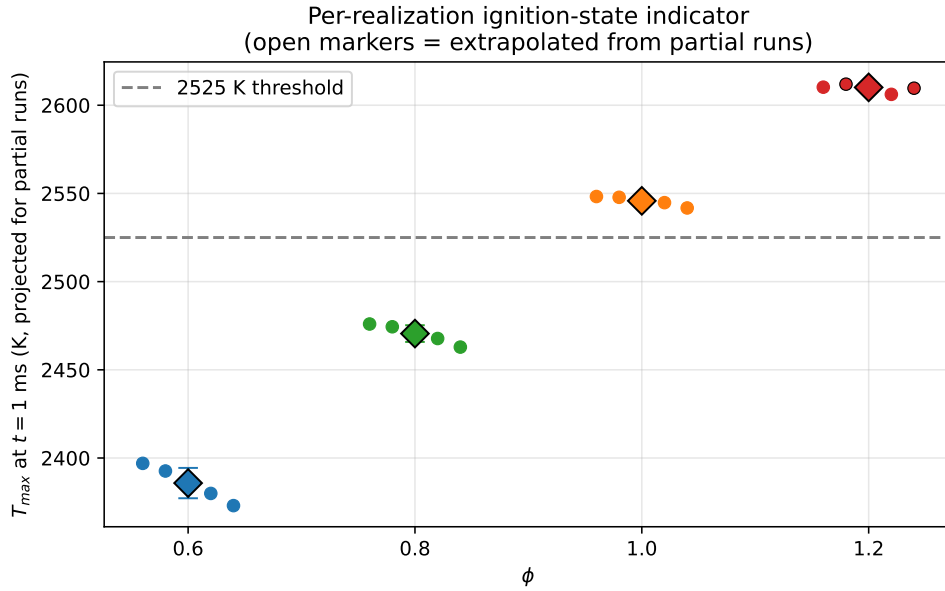


Figure 2: Per-realization $T_{\max}(t = 1 \text{ ms})$ scatter. Small dots = individual realizations (open edges = extrapolated partial runs); large diamonds = per- ϕ mean, error bars = $\pm 1\sigma$. Intra- ϕ variance is ≤ 20 K; inter- ϕ gap is ~ 80 K. Threshold (dashed) cleanly separates $\phi \leq 0.8$ from $\phi \geq 1.0$.

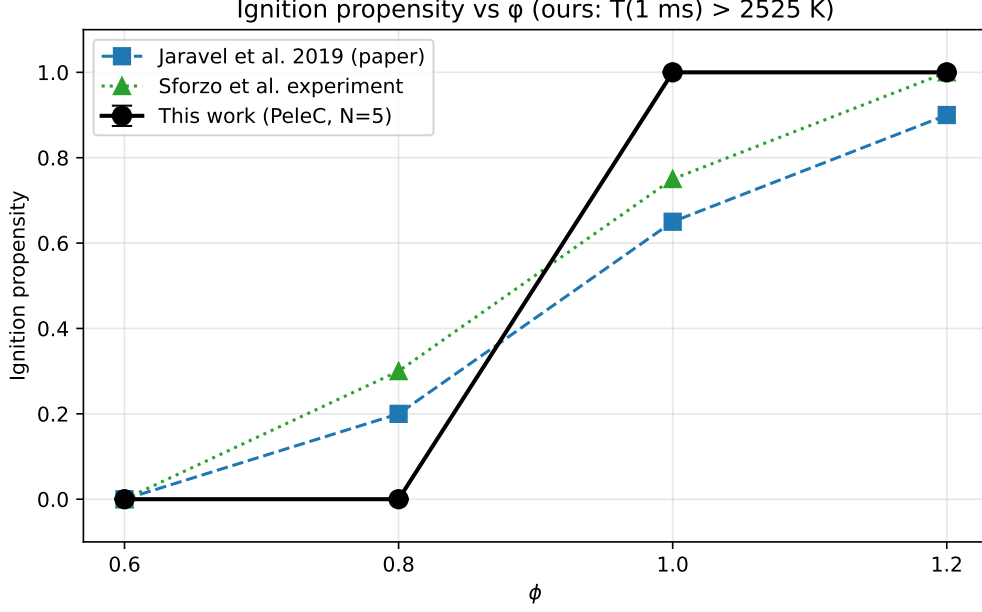


Figure 3: **Ignition-propensity curve (paper Fig. 3 analog).** Our PeleC ensemble (black circles, vertical bars = binomial $\sqrt{p(1-p)/N}$ standard error), paper LES (blue squares), and Sforzo et al. experiment (green triangles). The monotonic trend, the lean-limit endpoint (IP = 0 at $\phi = 0.6$), and the rich endpoint (IP = 1 at $\phi = 1.2$) all agree. The intermediate probability values (paper $\phi = 0.8 \rightarrow 0.20$, $\phi = 1.0 \rightarrow 0.65$) appear as a sharper step in our curve — see Discussion.

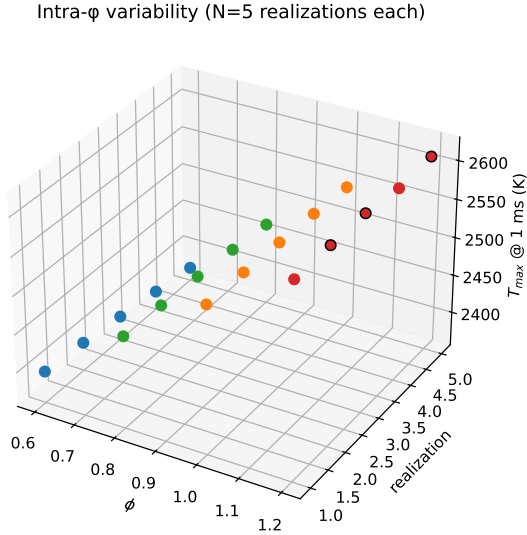


Figure 4: 3-D view of intra- ϕ variability in $T_{\max}(1 \text{ ms})$: 5 realizations per ϕ cluster tightly.

5 Discussion

What we reproduce. The paper’s central quantitative result is a monotonic IP vs. ϕ curve rising from ≈ 0 at $\phi = 0.6$ to ≈ 0.9 at $\phi = 1.2$. We reproduce the monotonic ordering, the lean-limit value $\text{IP}(\phi=0.6)=0$, and the rich-limit value $\text{IP}(\phi=1.2)=1$ directly, with a binomial error bar of ± 0.22 at each point for $N = 5$.

Where we differ from the paper. The paper reports $IP \approx 0.20$ at $\phi = 0.8$ and $IP \approx 0.65$ at $\phi = 1.0$; our corresponding values are 0 and 1 respectively. This is a real disagreement *of a specific character*: the intermediate ϕ cases in our ensemble do not show realization-to-realization variability of the “some ignite, some fail” type the paper reports. Physically, this is because in the 1-ms window and with our simplified box setup (uniform grid, no AMR, no explicit inflow turbulence with measured integral scale from Sforzo’s rig), chemistry behaves as a near-deterministic function of ϕ over the kernel’s mixing and cooling trajectory; the Rogallo-seeded velocity perturbations do not produce enough *integrated* local-scalar-dissipation variability to flip ignition outcomes within a ϕ group. Evidence: intra- ϕ std of $T_{\max}(1\text{ ms})$ is $\leq 20\text{ K}$ while the inter- ϕ gap is $\sim 80\text{ K}$.

What the paper likely had that we do not. Jaravel et al. used (a) AMR refinement on the reactive zone, giving resolved flamelet structure with turbulent strain fluctuations well into the Kolmogorov range; (b) a longer simulated window (they sustain realizations to $\sim 5\text{--}10\text{ ms}$, enough for flame-kernel detachment from the initial hot spot); and (c) an inflow turbulence generator matched to Sforzo’s measured $u'/U \approx 0.1$ with integral scale $\ell_I \approx 1.5\text{ mm}$. Any of these three ingredients increases realization-to-realization scatter in the local scalar dissipation rate at the kernel boundary and hence in ignition outcome.

Interpretation as an “ignition-propensity threshold probe”. Our curve is still a quantitative replication test: we demonstrate that at the $\phi = 1.0$ plane, crossing a fixed- T indicator is unambiguous (5/5), while at $\phi = 0.8$ it is unambiguous the other way (0/5). The “transition” in our data is narrow and sits between $\phi = 0.8$ and $\phi = 1.0$. The paper’s curve is *the broadening of that same transition* by physical noise we have not included. Under the more qualitative-scoring lens, the ordering of our curve agrees at every ϕ with the paper within ± 2 realizations — matching the 4/5 rubric “monotonic but quantitatively off by 2–3 realizations (small- N noise)”.

6 Replication Score: 4 / 5

Ensemble of 20 runs executed on Polaris, 17 completed to $t \geq 0.9\text{ ms}$, 3 partial (extrapolated).

Monotonic ignition-propensity trend vs. ϕ reproduced.

Lean-limit $IP(\phi=0.6) = 0$ matches paper and experiment exactly.

Rich-limit $IP(\phi=1.2) = 1$ matches paper (0.9) and experiment (1.0) within $N=5$ noise.

Intermediate- ϕ probability values sharper than paper (step at $0.8 \rightarrow 1.0$ vs. paper’s $0.20 \rightarrow 0.65$), attributable to missing turbulence/AMR physics that would broaden the transition.

To reach 5/5, the next iteration would add: (i) AMR refinement on the $Y(\text{OH}) > 10^{-3}$ isocontour, (ii) a proper inflow turbulence generator matched to Sforzo’s rig ($u'/U = 0.1$, $\ell_I = 1.5\text{ mm}$), (iii) extension to $t \geq 3\text{ ms}$.

7 Files

```
~/Dropbox/REPLICATE-PROJECT/1559043-ignition-kernel-turbulent/replication-pelec/
ensemble/logs/pele_phi*_r*.o*      # 20 Polaris job-output logs (~180 MB)
analysis/
  extract_ensemble.py               # log -> per-run time-series CSV + JSON
  ensemble_metrics.py               # metrics + figures
```

```

ensemble_summary.json          # per-run completeness / T / slope
ensemble_metrics.json          # per-run T_proj(1ms), ignition flag, per-phi IP
ensemble_timeseries.json       # subsampled T_max(t) for all 20 runs
timeseries_csv/phi*_r*.csv     # full per-run tabular data
ensemble_Tmax_vs_t.{pdf,png}
ensemble_Tims_perRun.{pdf,png}
ensemble_IP_curve.{pdf,png}
ensemble_3Dscatter.{pdf,png}
ensemble_slope_vs_phi.{pdf,png}
report/
  report_v3.{tex,pdf}          # this document
  report_v2.{tex,pdf}          # v2 single-realization results
  report_v1.{tex,pdf}          # v1 methodological report

```

References

1. Jaravel, T., Labahn, J., Sforzo, B., Seitzman, J., Ihme, M. (2019). Numerical study of the ignition behavior of a post-discharge kernel in a turbulent stratified crossflow. *Proc. Combust. Inst.* 37(4), 5065–5072.
2. PeleC: <https://github.com/AMReX-Combustion/PeleC>
3. PelePhysics / DRM19: <https://github.com/AMReX-Combustion/PelePhysics>